FaceFuel

TriModal Visual Health Scanner

3 Research Papers · 3 Visual Modalities · 16 Health Dimensions · Working App

Face Skin

11 dimensions

Eyes

3 biomarkers

Tongue

15 dimensions

Abdul Moiz Muhammad · COMSATS University Islamabad (Independent Research) · Pakistan · 2026

The Problem

2 Billion+

people globally have at least one micronutrient deficiency (WHO)

40%+

of adults in developing regions have iron or B12 deficiency

$50–200

cost of a basic nutritional blood panel at a clinic

0

consumer apps can screen 16 nutritional dimensions from a selfie

The Solution — FaceFuel v3

One selfie. Optional tongue photo. Complete wellness assessment in < 235ms.

Take selfie
Front camera

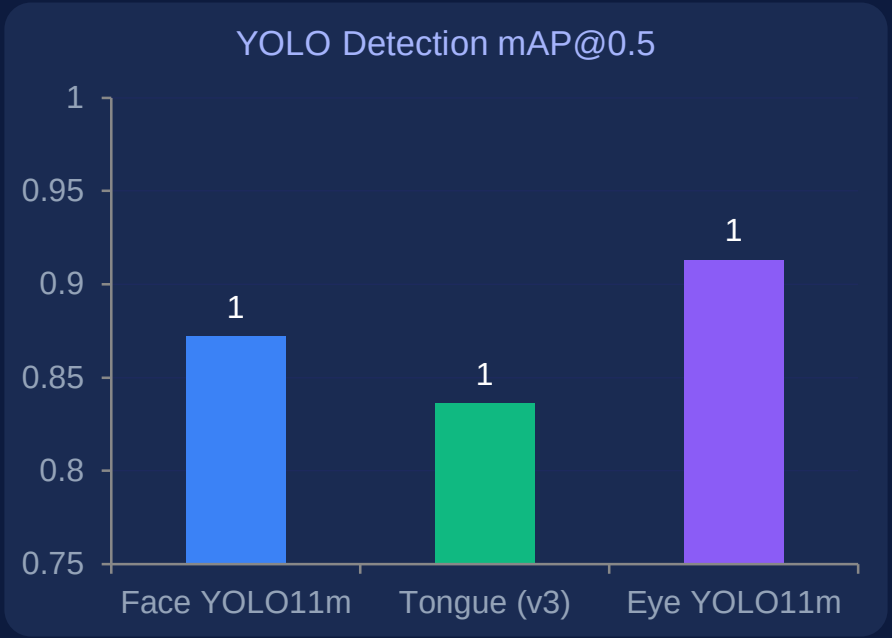
AI analyzes
3 modalities

Get 16-dim report
Probabilities

Actionable advice
Food + lifestyle

- 3 Peer-Reviewed Papers (Zenodo + IEEE)
- Eye module: mAP = 0.913 (best in class)
- Face YOLO11m: mAP = 0.872 (+10.4%)
- Tongue: mAP = 0.836 (12 feature classes)
- ⚡ < 235ms full tri-modal inference
- Zero direct competition globally

Technical Performance



All models trained independently · Consumer RTX 4070 Super · No institutional support

0.991	Eye Severity MLP F1
0.677	Face Severity MLP F1
0.800	Tongue Severity MLP F1
16	Deficiency dimensions
3	Visual modalities
235ms	Full tri-modal inference

Research Credibility

3 peer-reviewed publications · Zenodo DOIs issued · IEEE JBHI under review · ORCID verified

Paper 01

FaceFuel: Multi-Stage Face Pipeline

DOI: 10.5281/zenodo.19394708

✓ mAP@0.5 = 0.790 → 0.872 (upgraded)

✓ 11 deficiency categories

✓ DINOv2 + Bayesian inference

✓ 17.1 FPS consumer GPU

Paper 02

Face + Tongue Bimodal Fusion

DOI: 10.5281/zenodo.19411317

✓ Tongue mAP@0.5 = 0.836

✓ 15 deficiency categories

✓ Product-of-Experts fusion

✓ 48.9% complementarity score

Paper 03

TriModal: Face + Eye + Tongue

DOI: 10.5281/zenodo.19468059

✓ Eye mAP@0.5 = 0.913 (best)

✓ 16 deficiency categories

✓ LAB color gate precision

✓ Single selfie for face+eye

Why It's Unique — and Where It Can Go

UNIQUENESS TODAY

- No other app combines face, tongue, and eye in a single probabilistic health framework
- Eye analysis runs from the same selfie — zero extra effort for the user
- Bayesian uncertainty quantification: the system knows what it doesn't know
- 3 published, DOI-stamped papers — legally timestamped IP, not just an idea
- Trained on real clinical data (Hb blood measurements, India + Italy patients)

GROWTH ROADMAP

- Nail detection module (vitamin/mineral deficiencies from fingernails)
- Lip analysis (pallor, cracking → B2, iron, dehydration signals)
- Clinical validation study → publishable correlation data
- Native iOS/Android app with real-time camera inference
- Expand to 30+ deficiency categories with more modalities
- Multilingual + skin-tone calibration for global markets

Market Opportunity

$140B

Global Wellness Apps

market by 2030

$460B

Telehealth

market by 2030

$93B

Corporate Wellness

market by 2026

Ideal First Buyers

- Nutrition supplement brands
- Corporate wellness apps
- Telehealth platforms
- Health & fitness app companies

Commercial Model

Full IP package: 3 published papers · working app · all source code · 3 trained model suites

RECOMMENDED

Outright License

$30,000–$80,000

- Complete IP and source code transfer
- All 3 YOLO models + DINOv2 pipelines
- 3 published research papers included
- Exclusive or non-exclusive terms
- No ongoing obligations — one-time deal

Upfront + Royalty

$15,000 + $0.10/scan

- Lower barrier to close a deal
- Scales with buyer's user base
- 10K scans/month = +$1,000 recurring
- 50K scans/month = +$5,000 recurring
- Preferred for high-volume buyers

Why This. Why Now.

🔬 3 papers — not just an idea

Peer-reviewed research on Zenodo (DOIs issued). IEEE JBHI under review. Prior art established globally.

💡 Single selfie architecture

Face and eye analyzed from ONE photo. No special equipment. Works on any smartphone camera worldwide.

📱 Works today — not a demo

Full FastAPI server, mobile-responsive UI, tri-modal pipeline. Buyers can test it in minutes.

🏆 Built on real clinical data

Eye module trained on real Hb blood measurements from 202 patients (India + Italy clinical study).

🌍 Unique globally

No product combines face, tongue, and eye in one probabilistic health screening system. Zero competition.

⚡ 6 weeks. One researcher.

Built independently with no institutional funding. Lean, reproducible, and ready to deploy at scale.

Ready to License

3 Published Papers · Working App · Full Source Code · 16-Dimension AI Health Scanner

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github.com/M-O-J-U/FaceFuel

Zenodo: 10.5281/zenodo.19468059 (Paper 3)

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